Investigating the Impact of Losses on FGVC Performance and Geometry

Loss-function ablation and species-count scaling ladder on SWIN-Base

Internal draft: results so far

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Abstract

This report asks which training loss produces the best species classifier for digitised herbarium sheets, and whether the answer changes as the number of species grows. To isolate the effect of the loss, we keep everything else fixed: the same SWIN-Base backbone at 384 px with ImageNet-22k initialisation, the same 50-epoch training recipe, and the same Kaggle Herbarium 2022 data with 15,501 species. Only the classification head and its loss change between runs. We compare eight losses against a plain ArcFace reference. Every loss is trained to completion, and every model is scored on the full validation set of 167,955 images.

Two losses clearly win, and both work by giving each species its own margin instead of one shared margin. DBM beats ArcFace by +1.9 points of macro-F1 and +1.5 points of top-1 accuracy, and KappaFace by +1.8 and +1.2. Retraining with a second random seed moves each of these scores by about 0.1 points or less, so the gains are not noise. Most of the improvement comes from rare species. For species with 6–10 training images, per-species F1 rises by up to +6 points, while for species with 51–73 images it rises by only about +1. The two winners also split the work: KappaFace is better for species with fewer than about 10 training images, and DBM is better for species with more than about 15. Of the remaining losses, sub-center ArcFace gives a small but real gain, TARA has no detectable effect, ECC and HACE are slightly worse than ArcFace, and X2-Softmax fails because its features collapse. We also find that tidier average feature geometry does not mean a better classifier: the two best losses have looser species clusters than any loss in the ArcFace family. Finally, on a "scaling ladder" of nested label sets with 15.5k, 32k and 64k species, trained with the same compute at every size, DBM beats ArcFace on every rung. Its macro-F1 lead grows with the number of species, from +2.6 to +3.0 to +3.9 points, while its accuracy lead stays roughly flat at +1.0 to +1.3 points.

1 Motivation

Identifying the species on a digitised herbarium sheet is a fine-grained classification problem with a very long list of possible labels. Kaggle Herbarium 2022 alone contains 15,501 species, and the merged Herbarium 2021 and 2022 data contains 64,113 (de Lutio et al., 2021). Two properties of this data make the problem hard, and both shape how we evaluate the models in this report.

The first is that the remaining mistakes are between close relatives. When our SWIN-Base models get a specimen wrong, 75–78% of the time the species they predict is in the correct genus, and 87–90% of the time it is in the correct family (Section 6.2). If the wrong answers were random, these rates would be about 0.3% and 3%. In other words, the model has usually already worked out roughly where the specimen sits in the plant tree of life. What it gets wrong is the last step: telling apart sister species that may differ only in small details of the leaves, flowers or fruit on a pressed and dried sheet.

The second is that many species have very few examples. In Herbarium 2022, the number of training images per species ranges from 2 to 73. In the largest label set we use, with 64k species, the median species has only 10 training images and 3 validation images. This matters for the choice of metric. Top-1 accuracy counts every *image* equally, so it is dominated by the common species that contribute most of the images. Macro-F1 instead computes an F1 score for each species separately and then averages those scores, so every *species* counts equally, however rare it is. That makes macro-F1 the measure most sensitive to how well rare species are learned. We treat macro-F1 as our primary metric and report top-1 accuracy alongside it.

This report builds on an earlier controlled experiment at 224 px, which compared ordinary cross-entropy (CE) with ArcFace (Deng et al., 2019). ArcFace is a loss that forces the classifier to separate species by a fixed angular margin (Section 3 explains how). Switching from CE to ArcFace changed the structure of the feature space far more than it changed accuracy. We measure how well separated the species clusters are with a "separation gap" (defined in Section 6.3). That gap became 16 times wider, from 0.012 to 0.196, yet top-1 accuracy rose by only 1.7 points, from 0.7796 to 0.7970. So simply pushing species further apart in the ArcFace way did not seem to be what limits accuracy. That raised the two questions this report addresses. First, does a loss that intervenes somewhere else (through per-species margins, the taxonomy, an auxiliary loss term, or a pretrained teacher model) train a better fine-grained classifier? Second, does the answer still hold as the label set grows?

2 Fixed baseline: what stayed the same

The aim of the ablation is to attribute any difference in results to the loss and nothing else. Every run therefore uses the same backbone, the same data, the same optimiser and the same learning-rate schedule (Table 1). The only part of the model that changes is the final classification head and the loss computed from it.

The reference loss is ArcFace with one weight vector per species, a scale of $s = 30$ and a margin of $m = 0.30$ (both explained in the next section). It is applied directly to the 1,024-dimensional feature that SWIN produces for each image. There is no extra bottleneck layer, no BatchNorm, and no auxiliary cross-entropy head. We chose this configuration because its 224 px version is the one used in the earlier controlled CE-versus-ArcFace comparison, where it reached 0.7971 accuracy and 0.7605 macro-F1.

Moving from 224 px to 384 px forced one other change. Swin Transformers compute attention inside fixed-size windows of the image, and the window size is tied to the input resolution (Liu et al., 2021). A model pretrained for 224 px

Table 1: **Shared training recipe.** Identical for every loss in the ablation. The ladder runs (Section 5) use the same recipe with a fixed step budget in place of the epoch count.

Backbone	SWIN-Base, `microsoft/swin-base-patch4-window12-384-in22k`
Input	384×384, RandomResizedCrop + horizontal flip (no mixup, no advanced aug.)
Train / val data	Kaggle Herbarium 2022: 671,817 / 167,955 images
Label	`scientificNameEncoded`, 15,501 species (15,334 present in val)
Optimiser	AdamW (fused), lr 10^{-4}, weight decay 0.01
Schedule	500 warmup steps, cosine decay over 50 epochs (= 131,250 steps)
Batch	effective 256 (64×2 accum.$\times 2$ GPUs, or $64 \times 4 \times 1$ GPU)
Precision / seed	bf16 / 42 (seed 43 for the noise checks)
Head scale	$s = 30$ for every margin head
Evaluation	full validation set, top-1, top-5 and macro-F1 over present species
Hardware	1–2 NVIDIA L40S on BU SCC; 12 h jobs resumed from checkpoints

inputs therefore cannot be reused directly at 384 px. The 384 px runs use the `swin-base-patch4-window12-384-in22k` checkpoint instead, which was pre-trained for that resolution. At 384 px, ArcFace reaches 0.8299 accuracy and 0.7972 macro-F1, which is 3.3 and 3.7 points above ArcFace at 224 px. All comparisons between losses in this report are made at 384 px.

A correction to how 2-GPU runs were scored. Some losses were trained on two GPUs. In that setting, each GPU evaluates a different half of the validation set, and the two halves of the predictions have to be gathered together before the score is computed. The original training script never did this gathering step. As a result, every 2-GPU run reported its score on only the first GPU's half of the validation set. Accuracy was barely affected, but macro-F1 came out about 1 point too low, because with half the images each species has fewer validation examples. To remove this problem, we re-scored every finished model on the full validation set using a single GPU. For the models that had been trained on one GPU, this reproduced their logged numbers exactly, which confirms that the re-scoring is correct. All 15k-species numbers in this report are these full-set re-scores, and the scaling-ladder runs use the corrected code.

3 Background: how a cosine classifier head turns a feature into a prediction

Every loss in this study except HACE is built on the same kind of classification head, called a cosine head. Understanding it once makes each loss easy to describe as a small change to it, so we walk through it here.

The SWIN backbone turns each image into a vector of 1,024 numbers, which we call the image's *feature* $\mathbf{x}$. This is the model's learned summary of what it sees in the image. It helps to picture $\mathbf{x}$ as an arrow pointing in some direction in a 1,024-dimensional space.

The classification head stores one further vector of 1,024 numbers for each species

j. This is the species' *weight vector* $\mathbf{w}_j$. It is learned during training, and it acts like a template for the species: an arrow pointing roughly toward where the features of that species' images tend to land. With 15,501 species, the head holds 15,501 such vectors, stacked as the rows of a $15{,}501 \times 1{,}024$ weight matrix W.

To score an image against species j, the model compares the image's feature with that species' template. In an ordinary linear classifier, the comparison is a dot product plus a bias, $\mathbf{w}_j \cdot \mathbf{x} + b_j$. Doing this for every species at once is the matrix product $W\mathbf{x}$ (plus the bias vector), which gives one score per species. These scores are the *logits*, and a softmax turns them into probabilities.

A cosine head makes two changes to this. First, it rescales both $\mathbf{x}$ and every $\mathbf{w}_j$ to length 1 before comparing them, and it drops the bias. For two vectors of length 1, the dot product is exactly the cosine of the angle between them, which we write $\cos\theta_j$. The angle θ_j is therefore the natural measure of how well an image matches species j. An angle of 0° means the feature points exactly along the species' template, and an angle of 90° means the two are unrelated. Normalising means that only the *direction* of a feature matters, not its length. Second, because a cosine only ranges from -1 to 1, a softmax over raw cosines would give almost the same probability to every one of the 15,501 species. The head therefore multiplies every cosine by a fixed scale s, which we set to 30 for every head. The logit for species j is then $s\cos\theta_j$.

The *margin* losses below all keep this structure and change only one thing. During training, and only for the correct species y of each image, they replace the logit $s\cos\theta_y$ by something smaller. The logits of all the wrong species are left alone. Lowering the correct species' logit makes the training task deliberately harder. To still classify the image correctly, the model has to make its feature point even more closely toward its own species' template than would otherwise be necessary. That pulls each species' images together and pushes different species apart. At test time no margin is applied, and every head predicts with the plain logits $s\cos\theta_j$. The losses therefore differ only in how they shape the features and templates during training, not in how they make predictions afterwards.

4 The losses we ablated

The losses we test change training in three different places. Most replace the rule for the correct species' logit inside the cosine head. HACE changes the structure of the output, so that it predicts over the whole taxonomy rather than over species alone. ECC and TARA keep ArcFace's head and add an extra loss term on top of it. Table 2 summarises them. We chose this set to cover the main families of ideas for fine-grained and long-tailed recognition: fixed angular margins, margins that adapt per species, hierarchy-aware outputs, regularisers built on class centres, and alignment to a biology foundation model. A ninth planned method, FlexiGrad, changes how gradients from separate family, genus and species heads are combined. It needs those multi-task heads, which are not yet built, so it has not been run.

ArcFace (reference) (Deng et al., 2019). ArcFace applies the same margin to every species. During training, the correct species' logit becomes $s\cos(\theta_y + m)$ with $m = 0.30$ radians (about 17°). Adding m to the angle is equivalent to pretending

that the feature points 17° further away from its own template than it really does. For the image to still win, its true angle has to be small enough that even after this penalty the correct species beats every other species. Earlier ArcFace runs became numerically unstable when the full margin was applied from the start, so we ramp m linearly from 0 up to 0.30 over the first 5 epochs. We use ArcFace as the reference because it is the standard angular-margin loss and the strongest single change we had found before this study.

Sub-center ArcFace ($k = 3$) (Deng et al., 2020). Here each species has three weight vectors instead of one. An image is compared with all three, and the one it is closest to (the largest cosine) is used as that species' score. The margin rule is the same as in ArcFace. The motivation is that a single herbarium species can look quite different from sheet to sheet, for example when flowering versus fruiting, or when prepared in different ways. With several templates per species, each template can specialise in one of these appearances. The extra templates can also absorb mislabelled or unusual images without pulling the main template away from the typical ones.

KappaFace (Oinar et al., 2023). KappaFace keeps ArcFace's formula but gives each species its own margin, between 0 and a maximum of $m_0 = 0.30$. That margin is the average of two scores, each between 0 and 1. The first score depends on how many training images the species has. It is close to 1 for the rarest species and falls smoothly to 0 for the most common one. The second score depends on how tightly the species' features cluster. To measure this, the model keeps a running average of the normalised features of each species, updated at every training step. If a species' features all point in nearly the same direction, this average is a long vector; if they point in many different directions, they partly cancel and the average is short. From the length of the average, KappaFace estimates a concentration value κ (from the von Mises–Fisher distribution, a standard model for directions on a sphere), recomputed once per epoch. Species with a high κ, meaning a tight cluster, get a low score, and species with a spread-out cluster get a high one. The overall effect is that rare species and spread-out species receive margins close to 0.30, while common, tight species receive smaller margins. We included KappaFace because it targets the two difficulties of this data directly, few images and large variation within a species, without any per-species rule set by hand.

DBM, Difficulty-aware Balancing Margin (Son et al., 2025). DBM also gives each species its own margin, but builds it from two fixed parts. The *class* part depends only on the number of training images of the species, n_y: $m_C = K\,(n_y/n_{\min})^{-\tau}$, where $n_{\min}$ is the smallest image count of any species, and $K = 0.1$, $\tau = 1$ are the paper's defaults. With $\tau = 1$ this is simply $0.1 \cdot n_{\min}/n_y$. The rarest species get the largest margin, 0.1, and the margin shrinks in proportion to $1/n_y$ as species become more common. In Herbarium 2022, where $n_{\min} = 2$, a species with 73 training images gets a class margin of only about 0.003. The *instance* part is added only for "hard" training images, meaning images whose closest template (the species with the smallest angle) is not their own species. For these, the margin grows by $m_C(1 - \cos\theta_y)/2$, a fraction of the class margin that is larger the further

the image currently is from its own template. We included DBM because it was designed for long-tailed recognition, and it concentrates the training pressure on rare species and on images the model currently gets wrong, using nothing more than fixed class counts. It is worth noting that every DBM margin is far smaller than ArcFace's 0.30, and for common species it is close to zero.

X2-Softmax (Xu et al., 2023). X2-Softmax replaces the cosine for the correct species with a quadratic function of the angle, $f(\theta_y) = a(\theta_y - h)^2 + k$, and the logit becomes $s\, f(\theta_y)$. We use the paper's values $a = -1$, $h = -0.3$, $k = 1$. Like a cosine, this function is largest when the angle is zero and falls as the angle grows, but it falls at a different rate. The effect is a margin that changes with the angle instead of being a fixed number, which also removes the margin as a setting to tune. The idea was that a margin that adapts to how far each image is from its template could suit species of very different difficulty.

HACE, Hierarchy-Aware Cross-Entropy (Chan et al., 2026). HACE does not use a margin. Instead, it widens the output layer to cover the whole taxonomy tree: 15,501 species, 2,564 genera, 272 families and one root, 18,338 outputs in total. The probability of a genus, family or the root is computed as the sum of the probabilities of all the species beneath it, plus that node's own output. The training target is also spread up the tree. Instead of putting all of its weight on the correct species, the target gives part of it to that species' genus, a smaller part to its family, and so on, decreasing geometrically with a dilution factor of 0.6. The loss is cross-entropy between this spread target and the model's output over all nodes of the tree. Because most of our errors are already within the correct genus or family, we wanted to know whether a loss that knows about the taxonomy could make mistakes rarer, or at least milder.

ArcFace + ECC, Exploration of Class Centre (Yao et al., 2024). ECC keeps ArcFace and adds an extra loss term. For each species, it keeps a running average of that species' features, its "class centre". The extra term does two things. It pulls each image's feature toward its own species' centre, and it pushes the feature away from the centre of the most similar other species, pushing harder when the two species are more alike. It also builds a soft label for each species from the running average of the model's predicted probabilities for that species' images, and adds a term that keeps each image's prediction close to this soft label. The two parts are weighted equally ($\lambda_1 = \lambda_2 = 1$), because we could not recover the paper's own weights. We included ECC because it was validated on a Swin-Base backbone for fine-grained classification, the closest match to our setting among all the candidates.

ArcFace + TARA (He et al., 2026). TARA also keeps ArcFace and adds an extra loss term, this time using a frozen BioCLIP model (Stevens et al., 2024) as a teacher. BioCLIP is a vision foundation model trained on images from across the tree of life. For each training image, BioCLIP sees the same random crop as SWIN, resized to its own 224 px input. A small learnable projection layer maps SWIN's feature into BioCLIP's embedding space, and the extra loss rewards a high cosine similarity between the projected SWIN feature and BioCLIP's embedding.

Table 2: **Ablated losses and where they act.** n_y is the number of training images of species y. KappaFace and ECC keep per-species running statistics that are not synchronised across GPUs, so they train on one GPU, with the same effective batch.

Loss	Intervenes on	Margin on true class / extra term
ArcFace	head	fixed $m = 0.30$
Sub-center ArcFace	head	fixed $m = 0.30$, 3 centers per species
KappaFace	head	per species, ≤ 0.30, from n_y and feature concentration
DBM	head	per species $0.1\, n_{\min}/n_y$, plus extra for hard images
X2-Softmax	head	quadratic in angle, no margin setting
HACE	output structure	softmax over the full taxonomy tree
ArcFace + ECC	extra loss	pull to own center, push from most similar center
ArcFace + TARA	extra loss	cosine alignment with frozen BioCLIP

BioCLIP's weights never change. This is an adaptation of the original method, which aligns a large multimodal model rather than an image classifier. We included it because BioCLIP may already encode taxonomic structure that a classifier trained only on herbarium sheets would not learn by itself.

5 The scaling ladder

The loss ablation above uses a single label set of about 15.5k species. The per-species results (Section 6.2) suggest that DBM and KappaFace win mainly on rare species. Larger herbarium label sets mostly add rare species, so the natural next question is whether the ranking of losses holds as the number of species grows, and whether the gap between them widens. The scaling ladder is designed to answer this.

Building the label sets. We cut three label sets out of the merged Herbarium 2021 and 2022 data (Table 3). The smallest, called 15.5k, contains the 15,501 Herbarium 2022 species. The middle set, 32k, contains those same species plus 16,499 other species chosen at random. The largest, 64k, contains all 64,113 species. The sets are nested, so each larger set contains every species of the smaller ones, and each species keeps the same label number in every set it appears in. The validation images of each set are restricted to that set's species.

It is important to note that the 15.5k rung is not the same data as the 15k ablation, even though it has the same species. It includes the Herbarium 2021 images of those species as well, so it has about twice as many training images. It also uses a different validation split, and it is far more imbalanced, with up to 1,717 training images for a single species. Ladder numbers can therefore be compared with each other, across rungs, but not with the numbers in the 15k ablation tables.

Training protocol. We train DBM and ArcFace on every rung, six runs in total, using the recipe of Table 1. To keep the comparison across rungs fair in terms of compute, every run gets exactly the same budget: **131,250 optimizer steps at a batch size of 256**, which is the compute of one 15k ablation run. Because the larger rungs have more images, this budget covers fewer passes through their data: 24.6 epochs at 15.5k, 20.1 at 32k and 14.9 at 64k. ArcFace on the ladder uses the

Table 3: **Ladder label sets.** Nested sets from the merged Herbarium 2021 and 2022 manifest. Larger sets add mostly rare species: the median training images per species falls from 57 to 10.

Rung	Species	Train images	Val images (species in val)	Median images / species	
				train	val
15.5k	15,501	1,366,368	341,932 (15,501)	57	14
32k	32,000	1,667,994	417,176 (29,635)	20	6
64k	64,113	2,249,230	562,308 (57,094)	10	3

same code path as DBM, so the only difference between the two is the head. Each run is evaluated every 26,250 steps on the full validation set of its rung, with a single random seed. KappaFace runs on the ladder are in progress and are not reported here.

6 Results

6.1 Overall results on Herbarium 2022 (15.5k species)

Table 4 gives the final scores of every loss on the full validation set, and Figure 1 shows how far each loss is from ArcFace. To judge whether a gap is real, we use two separate checks, because there are two separate sources of uncertainty.

The first is uncertainty from which images happen to be in the validation set. We estimate it with a paired bootstrap. We draw a new validation set of the same size by sampling images with replacement, score both models on exactly the same draw, and record the difference between them. Repeating this 1,000 times gives a distribution of differences, and the middle 95% of it is the interval reported in brackets. If this interval does not contain zero, the gap is unlikely to be an accident of the validation images.

The second is uncertainty from training itself, since a different random seed gives a slightly different model. We measure it directly by retraining DBM, KappaFace and ArcFace with a second seed (43 instead of 42). The bootstrap interval does not include this seed-to-seed variation, so the two checks are complementary.

DBM and KappaFace win, and are nearly tied. Both beat ArcFace by 1.2–1.5 points of accuracy and 1.8–1.9 points of macro-F1. Changing the seed moves any of the three retrained losses by about 0.1 points or less, so these gaps are 20 to 40 times larger than the seed-to-seed variation. Between the two winners, DBM is ahead in both seeds, by 0.29 points of accuracy each time and by 0.11 and 0.14 points of macro-F1. On macro-F1 this difference is small: the bootstrap interval for KappaFace minus DBM, $[-0.36, +0.02]$, just includes zero, so we read the two as close to a tie on our main metric. DBM also has the best top-5 accuracy (0.962). It has a practical advantage as well: unlike KappaFace, whose per-species running statistics are not synchronised across GPUs, it can be trained on two GPUs.

Sub-center ArcFace gives a small but real gain, and TARA has no detectable effect. Sub-center ArcFace improves on ArcFace by 0.16 points of accu-

Table 4: **DBM and KappaFace are the clear winners at 15.5k species.** Final scores on the full Herbarium 2022 validation set (167,955 images, 15,334 species), after 50 epochs at 384 px. Seed 42 except where noted. Gaps to ArcFace are in points, with 95% paired image-bootstrap intervals in brackets.

Loss	Top-1	Macro-F1	Top-5	Δ Top-1	Δ Macro-F1
DBM	**0.8443**	**0.8161**	**0.9617**	+1.45 [1.30, 1.58]	+1.89 [1.70, 2.14]
KappaFace	0.8415	0.8150	0.9569	+1.16 [1.02, 1.29]	+1.77 [1.55, 1.95]
Sub-center ArcFace	0.8314	0.8003	0.9481	+0.16 [0.02, 0.29]	+0.31 [0.09, 0.50]
ArcFace + TARA	0.8305	0.7986	0.9470	+0.06 [−0.08, 0.20]	+0.13 [−0.10, 0.33]
ArcFace (reference)	0.8299	0.7972	0.9466	–	–
ArcFace + ECC	0.8261	0.7934	0.9434	−0.38 [−0.52, −0.24]	−0.38 [−0.63, −0.21]
HACE	0.8212	0.7908	0.9520	−0.87 [−1.02, −0.72]	−0.65 [−0.91, −0.45]
X2-Softmax	0.3782	0.3136	0.7700	–	–
Seed 43 reruns (noise check)					
DBM	0.8444	0.8159		vs. seed 42: +0.01 / −0.02	
KappaFace	0.8414	0.8145		vs. seed 42: −0.01 / −0.05	
ArcFace	0.8288	0.7976		vs. seed 42: −0.11 / +0.04	
Earlier reference points at 224 px (not directly comparable)					
CE baseline, 224 px	0.7796	0.7455	0.9328		
ArcFace, 224 px	0.7971	0.7605	0.9337		

racy and 0.31 points of macro-F1, and both intervals lie above zero. TARA's gains, 0.06 and 0.13 points, have intervals that include zero, so we cannot distinguish TARA from plain ArcFace.

ECC and HACE are worse than ArcFace. ECC loses 0.38 points on both metrics, and HACE loses 0.87 points of accuracy and 0.65 of macro-F1. HACE shows an interesting pattern, though. Its top-5 accuracy, 0.952, is higher than ArcFace's 0.947. In other words, HACE more often places the correct species among its five best guesses, but less often ranks it first.

X2-Softmax fails. X2-Softmax reaches only 38% accuracy. Its learning curve rose slowly throughout training and did not level off, so it was still learning, only far too slowly. A natural suspect was the scale: we used $s = 30$ for every head, while the X2-Softmax paper uses $s = 64$. A rerun at $s = 64$ learned even more slowly, reaching 0.112 accuracy at epoch 6.8 and 0.140 at epoch 10.2, and we stopped it there. The scale is therefore not the cause. The geometry analysis in Section 6.3 shows what goes wrong instead: its features collapse.

6.2 Per-species results and the role of training-set size

Overall macro-F1 is an average over all species, and an average can hide where a gain comes from. In this section we look at the F1 score of each species separately. A species' F1 is the harmonic mean of its precision (of the images the model labelled as this species, the fraction that really were) and its recall (of this species' images, the fraction the model found). We then group species by how many training images they have. Although Herbarium 2022 caps species at 73 training images, there is still a rare tail: 1,060 species have 10 or fewer. Figure 2a shows the mean per-species

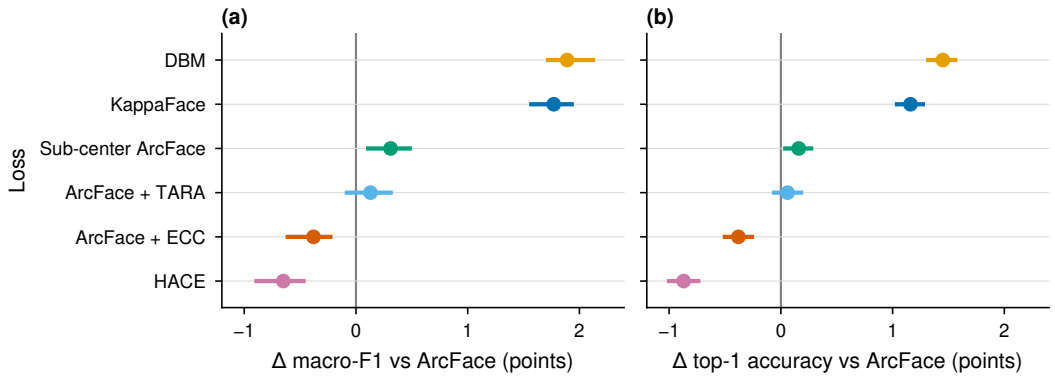


Figure 1: **Only the two per-species-margin losses beat ArcFace by more than a fraction of a point.** Each row is one loss; the dot is its score minus ArcFace's score, in percentage points, for (a) macro-F1 and (b) top-1 accuracy. A dot right of the vertical line means the loss beats ArcFace. The horizontal bar is the 95% interval from a paired bootstrap over validation images, so it reflects which images are in the validation set but not seed-to-seed variation; seed-to-seed variation, measured for DBM, KappaFace and ArcFace, is about 0.1 points or less. Full validation set, seed 42. X2-Softmax (−48 points of macro-F1, −45 of accuracy) is off the scale.

F1 of each loss in each group. Figure 2b shows the gain over ArcFace, computed species by species and then averaged within each group. Its intervals come from a bootstrap that resamples the *species* in each group (rather than the images), and again do not include seed variation.

The gains are concentrated in the rare tail. For species with 6–10 training images, KappaFace improves per-species F1 over ArcFace by 6.3 points and DBM by 4.6 points. For species with 51–73 images, which make up more than half of all species, both add only about 1 point (KappaFace 1.0, DBM 1.2). Sub-center ArcFace adds less than 1 point in every group. This matches what the two winning losses are designed to do, since both put larger margins on rare species.

Rare species remain much harder for every loss. Even the best loss in each group is about 24 points worse on species with 2–5 training images (KappaFace, 60.4) than on species with 51–73 (DBM, 84.6). The better losses narrow this gap but are far from closing it.

Fewer species are never recognised. A species has an F1 of zero when the model never predicts it correctly for any of its validation images. The number of such species falls from 367 with ArcFace to 313 with DBM and 255 with KappaFace.

No loss changes the kind of mistake. For every loss that trained properly, 75–78% of wrong answers still name a species in the correct genus, and 87–90% one in the correct family (for example, 75.5% and 87.7% for ArcFace, and 77.8% and 89.5% for DBM). The better losses therefore make fewer mistakes, but not different ones. Even HACE, which is built around the taxonomy, does not shift this pattern.

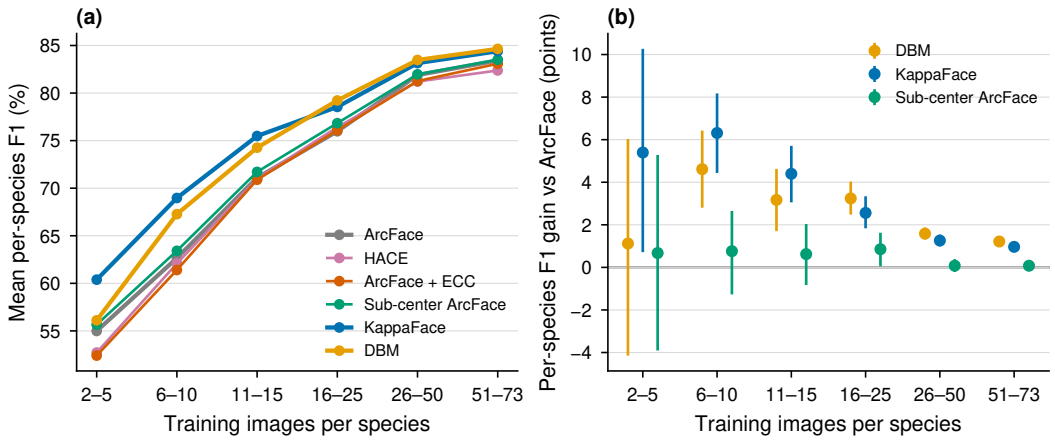


Figure 2: **DBM and KappaFace gain most on rare species; KappaFace leads on the rarest.** Species are grouped into bands by their number of training images; the bands hold 104 / 956 / 937 / 1,554 / 3,710 / 8,073 species, left to right. (a) For each loss, the average F1 score of the species in each band (ArcFace + TARA is omitted because it overlaps ArcFace). (b) For each species, the loss's F1 minus ArcFace's F1 on the same species, averaged within each band; above zero means the loss does better than ArcFace. Whiskers are 95% intervals from resampling the species in each band, so they are widest for the small 2–5 band, and they do not include seed-to-seed variation. Seed 42, full validation set; the 167 species with no validation images are excluded.

KappaFace and DBM win on different species. Figure 3 compares the two winners directly, species by species, using both seed pairs. KappaFace is better on species with fewer than about 10 training images: by 3.4 points for species with 2–5 images and 2.9 points for 6–10, with intervals above zero when both seeds are pooled. The two tie for species with 11–15 images. From 16 images upward, DBM is better, by 0.9, 0.4 and 0.2 points in the three remaining groups. These groups contain most of the species, which is why overall macro-F1 barely separates the two losses: KappaFace's large gains on a few rare species are offset by DBM's small gains on many common ones.

Combining the two winners without retraining. Because the two losses win on different species, we tested a simple way to combine them at prediction time. For each image, we look at the species KappaFace predicts. If that species has at most T training images, we keep KappaFace's answer; otherwise we use DBM's. To avoid tuning on the same images we report on, we chose T on a random half of the validation images ($T = 12$ for the seed-42 pair and $T = 10$ for the seed-43 pair) and scored the combined model on the other half (Table 5). This routing raises F1 on the rarest species by 3–8 points while leaving common species within ±0.1 points. Only 1.4–2.2% of images end up using KappaFace's answer, so the overall macro-F1 gain is small, +0.14 points, about the size of the seed-to-seed variation. The real benefit is for rare species, and the cost is running two models at prediction time.

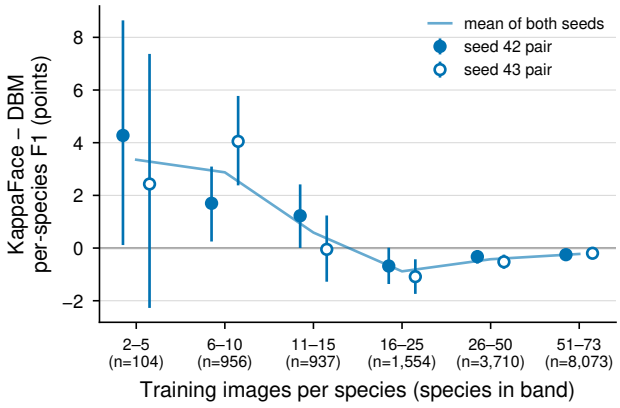


Figure 3: **KappaFace beats DBM on rare species, DBM wins on common ones.** For each species, KappaFace's F1 minus DBM's F1, averaged within each band of training images; above zero means KappaFace does better. The comparison is made twice, once between the two seed-42 models (filled) and once between the two seed-43 models (open). Whiskers are 95% intervals from resampling the species in each band; the line is the average of the two seed comparisons. The sign agrees across seeds in every band except 11–15, where the two losses tie.

Table 5: **Routing rare-species predictions to KappaFace lifts the tail.** Scores on the held-out validation half, with T chosen on the other half. Accuracy and macro-F1 are fractions; band columns are mean per-species F1.

Seed pair	Model	Top-1	Macro-F1	F1, 2–5 imgs	F1, 6–10 imgs
42 ($T = 12$)	DBM	0.8449	0.8078	0.517	0.672
	KappaFace	0.8426	0.8076	0.571	0.685
	Routed	**0.8456**	**0.8092**	**0.594**	**0.700**
43 ($T = 10$)	DBM	0.8458	0.8104	0.541	0.657
	KappaFace	0.8425	0.8061	0.566	0.686
	Routed	**0.8463**	**0.8118**	**0.591**	**0.708**

6.3 Representation geometry

Accuracy tells us how often a model is right, but not what its feature space looks like. To see whether the losses organise the feature space differently, we saved the 1,024-dimensional feature of every validation image for every model and computed the same geometry measures as in the earlier CE-versus-ArcFace study. All of them are based on each species' *centroid*, the average feature of its validation images.

The *within-species cosine* is the average cosine between an image's feature and its own species' centroid. Higher values mean that images of the same species point in more similar directions, so each species forms a tighter cluster. The *nearest-species cosine* is the cosine between each species' centroid and the closest centroid of any other species, averaged over species. Lower values mean that even the most similar neighbouring species is further away. The *separation gap* is the first minus the second. A large gap means clusters are tight compared with how close their nearest neighbour is. A gap near zero means a typical image sits no closer to its own species' centre than that centre sits to the nearest other species' centre. *NC1*

Table 6: **Geometry of the 384 px models** (full validation set, seed 42). The last column is the within-model Spearman correlation between per-species alignment and per-species recall.

Loss	Within	Nearest	Gap	NC1	Align.	ρ(align, recall)	Macro-F1
DBM	0.721	0.667	0.055	0.108	0.694	0.44	**0.8161**
KappaFace	0.733	0.622	0.111	0.089	0.763	0.45	0.8150
Sub-center ArcFace	0.773	0.532	0.241	0.063	0.795	0.38	0.8003
ArcFace + TARA	0.779	0.543	0.236	0.061	0.839	0.43	0.7986
ArcFace	0.772	0.530	0.242	0.062	0.838	0.44	0.7972
ArcFace + ECC	0.786	0.524	**0.261**	**0.056**	**0.840**	0.39	0.7934
HACE	0.684	0.661	0.023	0.142	0.474	0.46	0.7908
X2-Softmax	0.992	0.993	−0.001	0.017	0.610	−0.10	0.3136

compares the spread of features within each species with the spread between species centroids, so lower values again mean tighter, better separated clusters. Finally, the *alignment* is the cosine between each species' weight vector (its template in the classifier) and its feature centroid. High alignment means the classifier's template points straight at where that species' images actually land. By the usual reading, a higher gap, higher alignment and lower NC1 all indicate "tidier" geometry. Results are in Table 6 and Figure 4. The seed-43 retrains of DBM, KappaFace and ArcFace reproduce every value to within 0.003, so these measures are stable.

Across losses, tidier average geometry goes with lower macro-F1. This is the opposite of what one might expect. The two best losses, DBM and KappaFace, have the smallest separation gaps of any loss that trained properly apart from HACE: 0.055 and 0.111, against 0.242 for ArcFace. Their alignment is also lower, 0.69 and 0.76 against 0.84. In Figure 4a, they sit closer than any ArcFace-family loss to the diagonal, where a typical image is no closer to its own species' centre than that centre is to the nearest other species' centre. At the other end, ECC has the tidiest geometry of any loss on all three measures, yet the lowest macro-F1 of the ArcFace family. Combined with the earlier CE-to-ArcFace result, where geometry improved greatly but accuracy only a little, this tells us that average geometry is not a reliable target for choosing a loss on this data. One possible reading, which we have not yet tested, is that the averages are dominated by the many common species, whose clusters ArcFace-style losses make very neat, while DBM and KappaFace spend their effort on rare species instead.

Within a single model, geometry still predicts which species are recognised. Although geometry does not rank the losses, it remains informative inside each model. For every loss, species whose template is better aligned with their cluster have higher recall. The Spearman rank correlation between per-species alignment and per-species recall is 0.38–0.46 across models (last column of Table 6).

X2-Softmax's features have collapsed. For X2-Softmax, the within-species and nearest-species cosines are both about 0.99. Almost every image's feature points in nearly the same direction, regardless of species, so the model has to separate 15,501

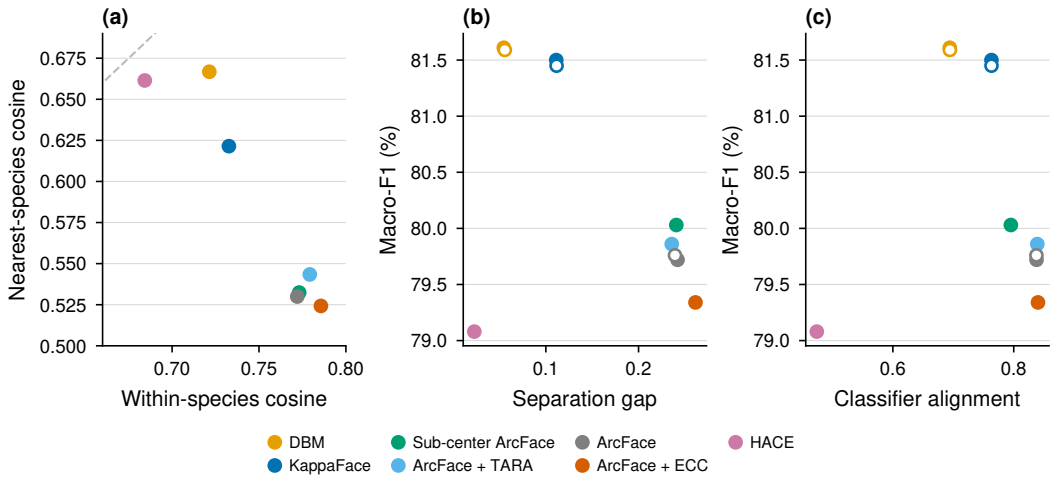


Figure 4: **The two best losses have looser geometry than every ArcFace-family loss.** Each point is one model, with every measure averaged over the 15,334 validation species. (a) Horizontal axis: how tight each species' cloud is, the average cosine between an image's feature and its own species' centroid (higher is tighter). Vertical axis: how close the nearest other species is, the cosine between each species' centroid and the closest other centroid (higher is closer). On the dashed diagonal the two are equal, meaning a typical image sits no closer to its own species' centre than that centre sits to its neighbour's centre. Points further below the diagonal have tighter clouds with more distant neighbours. DBM and KappaFace sit closer to the diagonal than any ArcFace-family loss; only HACE is closer. (b) Macro-F1 against the separation gap, the horizontal minus the vertical value in (a). (c) Macro-F1 against alignment, the average cosine between each species' classifier weight vector and its centroid. Filled markers are seed 42; open markers are the seed-43 retrains, which land on top of them. X2-Softmax (macro-F1 31%, collapsed features) is off the scale.

species inside a very narrow cone of directions. This is consistent with how slowly and poorly it learned.

HACE's species templates are poorly aligned. HACE has the lowest alignment of the working losses, 0.47. This needs care in interpretation: HACE's classifier also has outputs for genera and families, and its prediction combines information across the whole tree, so its species templates alone describe only part of the model.

6.4 Scaling ladder

Table 7 and Figure 5 give the results of the ladder.

The ranking holds at every size. DBM beats ArcFace on both accuracy and macro-F1 on all three rungs.

DBM's macro-F1 lead grows with the number of species, but its accuracy lead does not. DBM's macro-F1 lead increases from 2.6 points at 15.5k to 3.0 at

Table 7: **DBM beats ArcFace on every rung, and its macro-F1 lead grows with the number of species.** Final scores after 131,250 steps, full validation set of each rung, one seed per run.

	DBM		ArcFace		DBM lead (points)	
Rung (epochs)	Top-1	Macro-F1	Top-1	Macro-F1	Top-1	Macro-F1
15.5k (24.6)	0.8959	0.8436	0.8842	0.8178	+1.17	+2.58
32k (20.1)	0.8882	0.7754	0.8777	0.7452	+1.05	+3.02
64k (14.9)	0.8619	0.6818	0.8490	0.6429	+1.29	+3.89

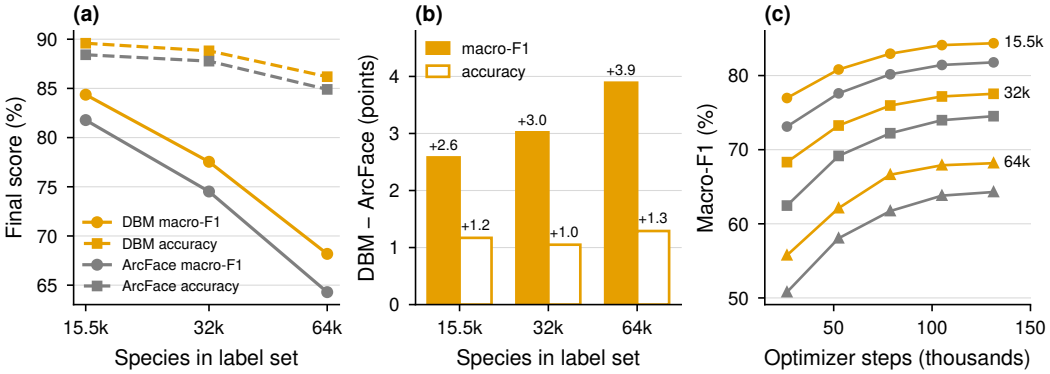


Figure 5: **DBM's macro-F1 advantage grows as rare species are added.** (a) Final macro-F1 (solid lines) and top-1 accuracy (dashed lines) of DBM (orange) and ArcFace (grey) on the three nested label sets; the horizontal axis is logarithmic in the number of species. (b) DBM's score minus ArcFace's score at the end of training, in percentage points, for macro-F1 (filled bars) and accuracy (open bars). (c) Macro-F1 at each of the five evaluations during training, one every 26,250 steps; colours as in (a), with one marker shape per label set, labelled at the right. Every run trains for the same 131,250 steps, so larger label sets get fewer epochs. Single seed per run; seed-to-seed variation at 15k was about 0.1 points.

32k and 3.9 at 64k, while its accuracy lead stays roughly flat at 1.0–1.3 points. The explanation follows from how the rungs are built and how the two metrics weight species. The larger rungs mostly add rare species (Table 3): the median number of training images per species falls from 57 to 20 to 10. Rare species contribute few images, so they barely affect accuracy, but each one counts as much as any other species in macro-F1. This matches the 15k per-species result, where DBM's gains were largest for rare species.

Adding species costs much more macro-F1 than accuracy. Going from 15.5k to 64k species, DBM's macro-F1 drops from 0.844 to 0.682, while its accuracy drops only from 0.896 to 0.862. ArcFace shows the same pattern. The species added at each step are hard to learn from so few images, but because they have few images they hardly register in accuracy.

The lead appears early and narrows slowly. At the first evaluation, after 20% of training, DBM already leads by 3.9, 5.9 and 5.0 points of macro-F1 on the

three rungs. The lead shrinks during training but never closes (Figure 5c). Both losses are still improving slowly at the end of training, most visibly on the 64k rung, which gets the fewest passes over its data.

Caveats. Each ladder point comes from a single seed. The seed-to-seed variation we measured at 15k was about 0.1 points, far below these gaps, but it was measured on different data. Every rung gets the same compute, so the larger rungs are trained for fewer epochs, and with more training part of the gap between rungs might close. Finally, on the 64k rung the median species has only 3 validation images, so the F1 score of any individual species there is very noisy, even though the average over tens of thousands of species is stable.

7 Summary of findings so far

The clearest result is that losses which give each species its own margin, DBM and KappaFace, are the only interventions that improve on ArcFace by more than a fraction of a point. They add about 1.9 points of macro-F1 and 1.2–1.5 points of accuracy, and the gain holds with a second seed. Looking per species shows that these gains are concentrated in the rare tail, and that the two losses divide it between them: KappaFace is better below about 10 training images and DBM above about 15. A simple rule that routes rare-species predictions to KappaFace, with no retraining, lifts the rarest species by 3–8 points.

The other ideas we tested did not help as implemented. Using the taxonomy (HACE), pulling features toward class centres (ECC) and aligning with BioCLIP (TARA) all matched or fell below ArcFace, and no loss changed the share of errors that land in the correct genus, which stays at 75–78%. The geometry analysis adds a caution: across losses, tidier average geometry did not mean higher accuracy, and the best losses have looser species clusters than any ArcFace-family loss. Finally, on the scaling ladder DBM beats ArcFace at 15.5k, 32k and 64k species, and its macro-F1 lead grows as more, mostly rare, species are added.

Not yet measured. Several pieces are still missing. KappaFace runs on the ladder, and the routing test on the ladder, are in progress. Per-species F1 by training-image count on the ladder rungs needs the per-image predictions of the ladder models to be saved, which has not been done yet. Geometry has not yet been split by rarity group, which would test the reading offered in Section 6.3. FlexiGrad and the multi-task family, genus and species heads have not been built. And apart from DBM, KappaFace and ArcFace, every loss has been run with only one seed.

A Provenance

Figures are generated by `figs/make_figures.py` from the data files in `figs/data/`.

Table 8: Run identifiers behind each result (first 8 characters).

Result	Runs
Full-set re-scores of the 8 losses	`f701e70f`
Training runs (seed 42)	ArcFace `6a899d4a`, sub-center `47362b64`, HACE `677aa821`, DBM `c2fecdb6`, X2 `0b103f18`, KappaFace `f13e6236`, ECC `8fff330a`, TARA `601daea1`
Seed-43 reruns	DBM `61e3b0a7` (re-score), KappaFace `cf97fa8d`, ArcFace `f4e0099f`
Per-species F1, bootstrap, error structure	`2964fe72`
KappaFace–DBM crossover and routing	`ab4f160f`
Geometry (11 models)	`bf50bd59`
X2-Softmax at $s = 64$	`877404ac`
224 px references	CE `be436c7e`, ArcFace `be730cdf` / `da0667dc`
Ladder data build	`5c9db6fa`
Ladder final runs (DBM / ArcFace)	15.5k `dc59af2e` / `e3ecf637`, 32k `ca0355e1` / `f30277ed`, 64k `29abd9c9` / `2244e684`

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